

Seismic behavior of a full-scale RC frame repaired using CFRP laminates

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Abstract

The opportunities provided by the use of Carbon Fiber Reinforced Polymer (CFRP) composites for the seismic repair of reinforced concrete (RC) structures were assessed on a full-scale dual system subjected to pseudodynamic tests in the ELSA laboratory. The aim of the CFRP repair was to recover the structural properties that the frame had before the seismic actions by providing both columns and joints with more deformation capacity. The repair was characterized by a selection of different fiber textures depending on the main mechanism controlling each component. The driving principles in the design of the CFRP repair and the outcomes of the experimental tests are presented in the paper. Comparisons between original and repaired structures are discussed in terms of global and local performance. In addition to the validation of the proposed technique, the experimental results will represent a reference database for the development of design criteria for the seismic repair of RC frames using composite materials.

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1. Introduction

A full-scale RC dual-system (Fig. 1) was subjected to unidirectional pseudodynamic tests in the ELSA laboratory of the Join Research Center (JRC) in Ispra (Italy). The structure consisted of two frames, one designed according to Eurocode 8 (i.e., EC8 frame) and the other according to the deformation-based design method proposed by Fardis and Panagiotakos [1] (i.e., DbD frame). Two tests were performed on the original structure: the first under the design earthquake and the second under the design earthquake times 1.5 [2]. After the damage, the structure was repaired using Carbon FRP (CFRP) laminates and then subjected to a new series of two tests with the same input accelerograms selected for the original. The laboratory activity was

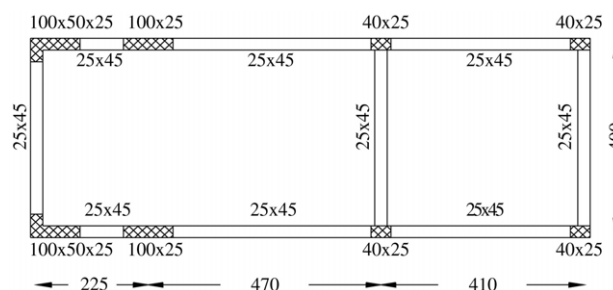


Fig. 1. Floor plan view of the structure (dimensions in cm).

concluded by a cyclic test up to the onset of structural failure to determine the ultimate capacity of the system.

The experimental program aimed at two main objectives: to compare the response of frames designed according to different approaches and to assess the opportunity of using composite materials as an effective technique for the seismic repair of RC frames. The discussion of the former is out of the scope of this paper; relevant details can be found

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in [2]. The latter is herein discussed presenting the reasons and the basis of the design of the CFRP repair, focusing on the comparison between experimental performance of original and repaired structure, and identifying opportunities and limitations of the adopted repair criteria as emerged by the tests on the full-scale structure.

2. Upgrade and repair of earthquake resistant structures

Different strategies can be pursued in the seismic upgrade and repair of existing structures; a first selection among available opportunities is generally made based on the deficiencies that the analysis or the observed damage of the building has pointed out. Then, cost–benefits considerations, ease of construction and evaluations on the impact that the upgrade and/or the repair can have in terms of loss of space and/or functions suggest the best solution among those possible. Seismic guidelines, such as FEMA 356 [3], classify the strategies for seismic rehabilitation as: (a) Local modification of components, (b) Removal or lessening of existing irregularities and discontinuities, (c) Global structural stiffening, (d) Global structural strengthening, (e) Mass reduction, (f) Seismic isolation, (g) Supplemental energy dissipation.

The present paper focuses on the local modification of components, which was the strategy adopted in the full-scale tests described in the following sections. The local modification of components is based on the upgrade of single members characterized by unsatisfactory seismic performance. The actions of this strengthening strategy can be of three types: (1) increase of ductility of elements already characterized by flexural collapse; (2) modification of collapse mode of elements from shear (brittle) to flexure (ductile); (3) increase of column strength capacity in order to modify the hierarchy of strength of the structure. The first action can be performed increasing the confinement level of elements in order to increase the strain capacity of concrete and fully exploit the steel yielding. The second action can be performed by increasing the shear strength of the element in order to make it larger than the flexural capacity at yielding and then avoid a brittle behavior. The last action can be carried out by increasing the strength of some elements (i.e., columns and joints); in this way, the failure of others (i.e., beams) occurs before and then prevents that of the upgraded members. This third choice allows protecting those elements whose failure could be critical from a global standpoint and then improving the seismic behavior of the structure.

The final goal of the mentioned actions is to move the structure toward ductile and energy dissipating mechanisms under seismic events; such concept is a key criterion in the seismic design and, in the last few decades, it has been adopted by seismic codes of Europe, USA, New Zealand and Japan.

Different techniques have been developed in order to achieve local modification of structural components.

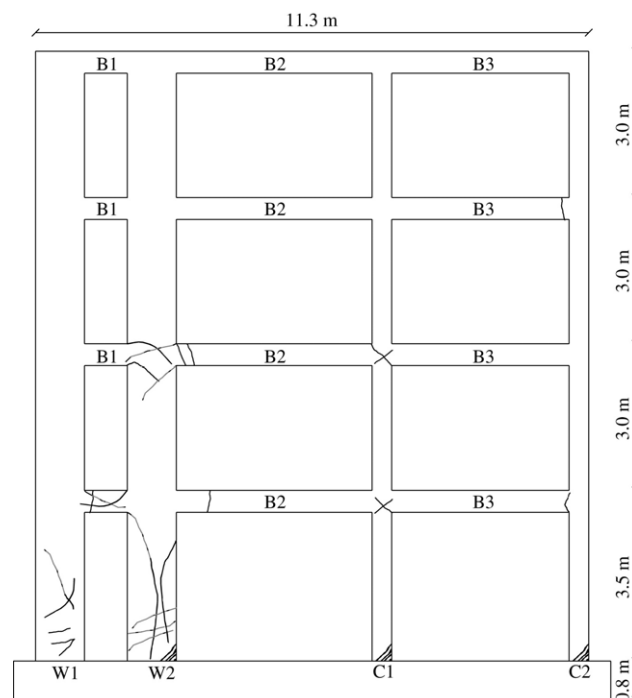


Fig. 2. Lateral view of the damaged structure.

Reinforced concrete jacketing, steel profile jacketing and steel encasement have been widely used in the past. All of them were characterized by disadvantages related to constructability (i.e., difficulty of ensuring perfect bond and collaboration between old and new parts, loss of space, construction time and high impact on building functions) and durability issues; in the case of reinforcing concrete jacketing, significant mass increase could also be generated. Innovative techniques based on FRP materials appear to be interesting alternatives to those solutions; along with high structural effectiveness, composite materials are light and easy to install, their application does not imply loss of space and, in some cases, it can be performed without interrupting the use of the structure. Antonopoulos and Triantafillou [4], Pantelides et al. [5], and Prota et al. [6] provided experimental validation of different FRP-based techniques used for repair and strengthening RC members. Advantages and limitations of using FRP composites in seismic applications were outlined by Manfredi and Prota [7].

3. Geometry, material properties and test set-up

The structure consisted of two geometrically identical frames connected by means of a rigid RC slab 150 mm deep. Each frame comprised two coupled shear walls (one L shaped and one rectangular) and two columns with beams connecting them. The dimensions of elements and structure are shown in Figs. 1 and 2. Design material properties were concrete strength equal to 25 MPa and steel yielding stress of 500 MPa [2]. With respect to FRP laminates, for the two lower floors quadriaxial FRP laminates with density

of 0.76 kg/m^2 and biaxial/uniaxial with density equal to 0.60 kg/m^2 were used; for third and fourth floors, fiber densities were 0.38 kg/m^2 and 0.30 kg/m^2 , respectively.

The lateral forces were imposed by means of hydraulic actuators (two per floor) placed against a reaction wall. Absolute rotations at different points of the structure were recorded by 96 inclinometers. Displacement transducers were placed at the base of all columns of the first level, on the shear walls (vertical and diagonal) and on the coupling beams.

4. Design of the repair using CFRP laminates

The original frame was first subjected to the design earthquake that caused only small cracks; the 1.5 design earthquake was then applied and the test was stopped as soon as important damage was observed (see Fig. 2 where lines and filled areas indicate cracks and concrete spalling, respectively). The failure of the original structure was due to shear collapse of some shear walls, due to insufficient anchor length in the horizontal bars connecting the confined core at the sides of the walls [2].

Once testing of the original structure was completed, the cracks caused by the 1.5 design earthquake were epoxy injected. Then, the structure was strengthened by using CFRP laminates with uniaxial, biaxial (0° – 90°) and quadriaxial (0° – 90° – $\pm 45^\circ$) fiber texture. The design of the repair and the selection of fiber textures on different structural elements were performed taking into account damages and deficiencies underlined by tests on the original configuration, and aiming at optimizing the benefits of the externally bonded FRP reinforcement along the directions of dominant stresses. The bases of the design of the CFRP repair are outlined in the following sections with reference to joints, columns, walls and beams.

4.1. Beam–column/wall joints

The shear improvement provided by FRP laminates was assessed according to the approach proposed by Antonopoulos and Triantafillou [8] that, based on equilibrium considerations, allows following the possible states of the joint behavior up to failure. Once geometric, bond and material properties are given and the acting axial forces are evaluated, the equations provide the inclination of the principal tensile stress, θ , and the shear stress, ν , corresponding to any given state of joint strains. The failure of the FRP strengthened joint occurs when either concrete crushes (i.e., the principal compressive stress attains the crushing strength of concrete) or FRP fails (i.e., the ultimate stress is attained or debonding occurs).

The design was targeted in order to quantify the amount of FRP needed on the joints in order to increase by at least 25% their original shear strength; such an amount was found to be equal to two plies of FRP laminates. Checks were performed for all joints (i.e., each column/wall

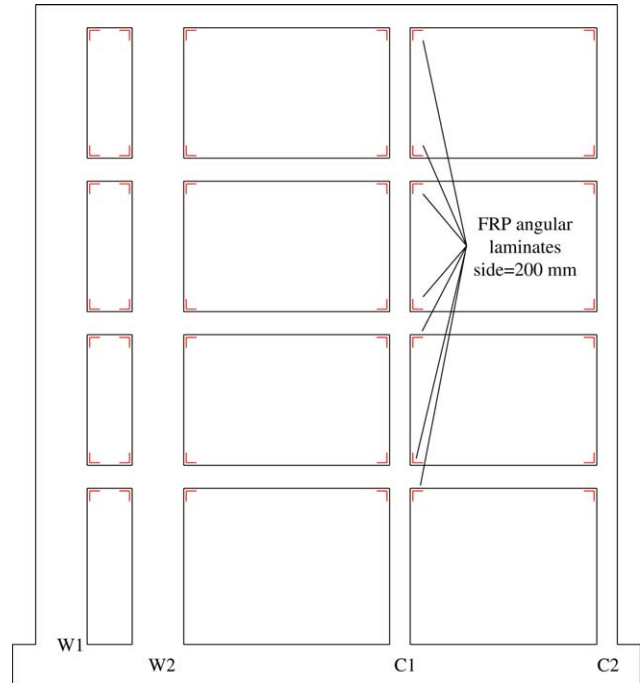


Fig. 3. CFRP angular laminates.

at each floor) in order to verify that an increase of at least 25% was achieved with that amount of FRP material. The theoretical failure mode was always concrete crushing, provided that proper anchorage would be ensured to prevent FRP debonding. As an example, for the joint on column C2 (Fig. 2) at the first floor, it was computed that two plies of CFRP bidirectional laminates with density equal to 0.60 kg/m^2 could increase its shear strength by 26.3% (i.e., 7.19 MPa versus 5.69 MPa) with a change in slope of principal tensile stresses from 39.97° to 41.79° .

Prior to laminate installation on the joints, FRP L-shaped laminates were placed to improve the diffusion of stresses between members (Fig. 3). Then, as mentioned, a total of two plies of fibers were installed on each joint; in order to ensure a proper anchorage, the former was extended for 200 mm on each side of the column (Fig. 4) and the latter for the same length on both sides of the beam (Fig. 5). A quadriaxial fiber texture was adopted on the joints of both walls and of the interior column (i.e., W1, W2 and C1), while a biaxial texture was selected for the exterior column C2 (Figs. 4 and 5). Fiber texture was differentiated for the two columns with the objective of comparing the performance of the two solutions. Tests conducted by Antonopoulos and Triantafillou [4] and Prota et al. [9] showed the effectiveness of such a solution in order to increase both strength and ductility of exterior and interior joints, respectively.

4.2. Columns and walls

The analysis of the behavior of the original structure suggested that it could be beneficial to provide the concrete of both columns and walls with a higher compressive

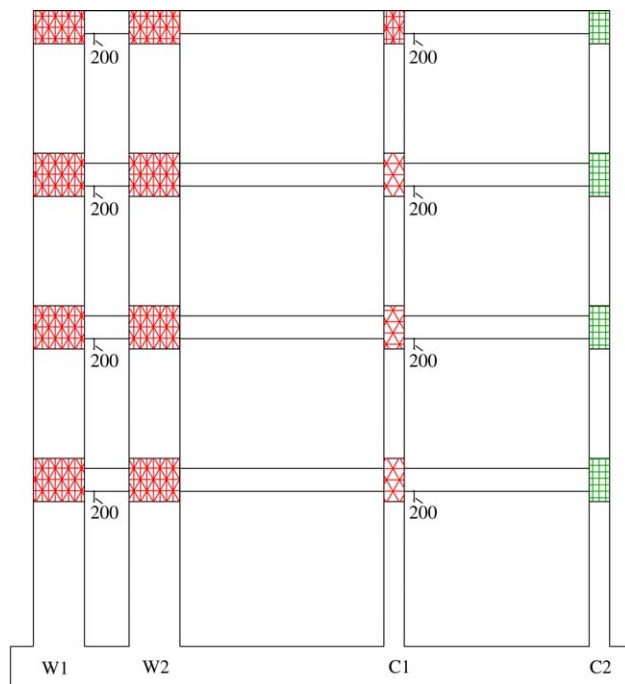


Fig. 4. Vertical CFRP laminates along walls and columns.

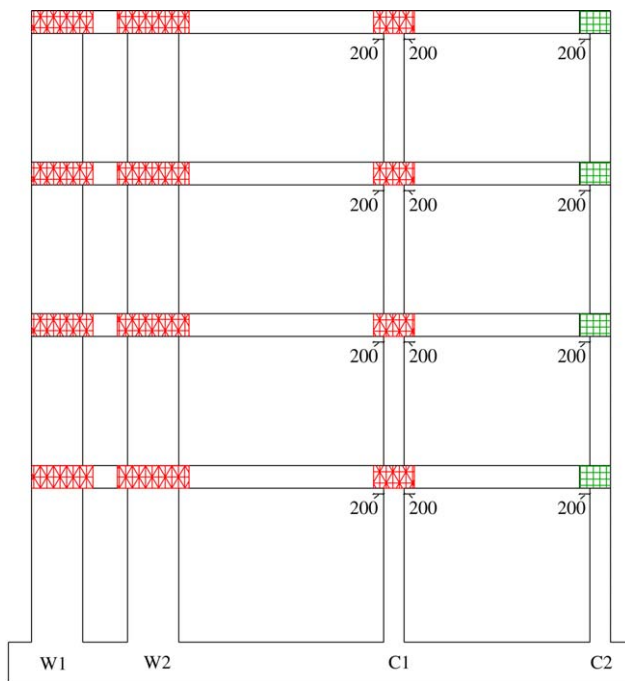


Fig. 5. Horizontal CFRP laminates along beams.

deformation capacity with the aim of restoring the global performance attainable prior to the damage. When such an objective is pursued by using FRP laminates, uniaxial fabrics are generally selected and wrapped around columns in the direction parallel to internal hoops [10,6,11]. Tests performed by Parretti and Nanni [12] showed that satisfactory results could also be achieved by multi-axial texture with a certain orientation on the hoop direction; even

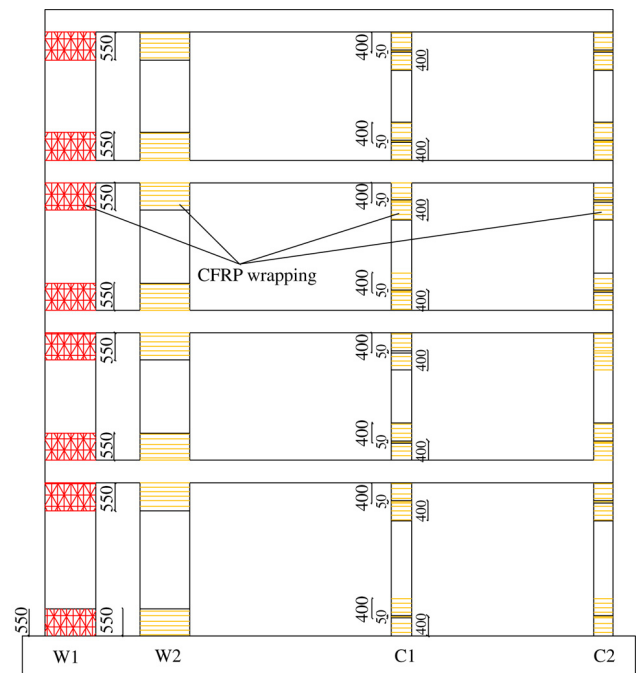


Fig. 6. CFRP wrapping of both walls and columns.

though such solution provided strength values lower than fibers parallel to the hoops, it allowed for a more ductile behavior in terms of both deformability and failure mode.

Both columns and walls of the tested frame were wrapped with CFRP laminates; the design of the FRP jacketing was based on theoretical analyses in terms of P – M interaction diagrams as well as moment–curvature relationships [13]. The effect of the CFRP wrapping was estimated by considering an axial load ratio of columns ranging between 0.05 (higher levels) and 0.12 (lower levels); these values were computed based on axial loads due to dead loads plus combined live loads. Calculations showed that, within that range of axial load ratios, the use of one ply of CFRP wrapping could boost column strength by 5%–10%, whereas significant increases (larger than 100%) of the cross section ductility (i.e., ultimate curvature) were expected.

The wrapping was extended for 550 and 750 mm along wall and column height, respectively (Fig. 6). In order to compare the benefits provided by multi-axial versus uniaxial fibers, a quadri-axial texture was selected for wall W1, while uniaxial fibers (parallel to hoop direction) were wrapped around the other vertical members (Fig. 6). At the base of vertical elements of the first floor, a steel strip was bonded after FRP wrapping installation [14].

In addition to the above mentioned confinement at top and bottom of both walls, further external FRP reinforcement was designed for them accounting for the fact that under the 1.5 design earthquake they experienced a shear failure highlighted by a multidirectional crack pattern (Fig. 7). Experimental tests [15] demonstrated the effectiveness of externally bonded FRP laminates in order to increase the shear capacity of deficient walls and achieve a stable



Fig. 7. Repaired cracks on the walls.



Fig. 8. Walls at the first floor.

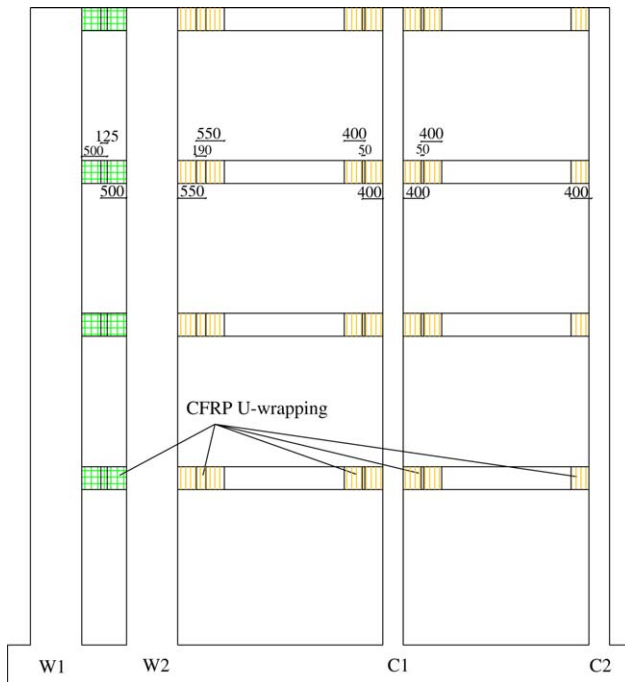


Fig. 9. CFRP U-wrapping of beams.

hysteretic behavior with high ductility levels. Considering the specific state of stress, the shear upgrade of both walls was achieved by wrapping along its entire height quadriaxial laminates (Fig. 8).

4.3. Beams

No FRP repair was designed for long beams (between W1 and C1, C1 and C2 in Fig. 9); however, a U-wrapping with uniaxial fibers parallel to the column axis was installed at their ends (Fig. 10); this was done to ensure a good anchorage for the joint laminates without increasing the strength of the end zone of the beams under bending, and without modifying the strength hierarchy of the nodal zone.

For short beams (between W1 and W2 in Fig. 9), whose behavior was controlled by shear more than bending, the strengthening was based on quadriaxial laminates that could provide the most appropriate response to the acting state of stress in order to increase both shear strength and ductility of the members. Laminates were installed along the entire length of horizontal members at each level (Fig. 8).

The weight of FRP materials used for the repair of the whole frame (Fig. 11) was estimated equal to 0.8% of that of the original RC structure.

5. Tests on the repaired structure

No major damage was observed during the application of the design earthquake. Horizontal cracks at the base of the shear walls were noticed; the cracks were mainly in the foundation elements or exactly at the cross-section between the foundation and the vertical elements. This underlined the beneficial effect provided by the CFRP upgrade that



Fig. 10. U-wrapping of beams.

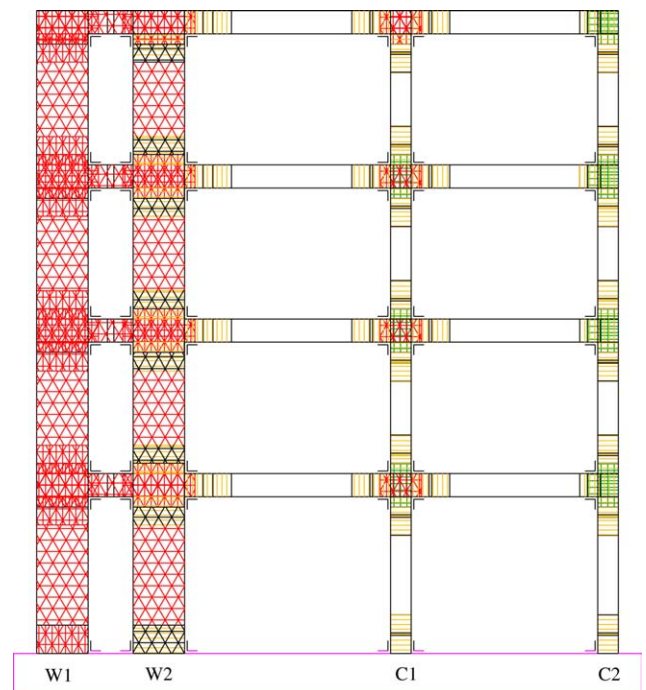


Fig. 11. Final scheme with complete wrapping of both walls.

allowed keeping the integrity of the part of structure above the foundation. Similarly to what happened for the original frame, the application of the 1.5 design earthquake caused a more significant damage on the repaired structure. The vertical rebars of the shear walls fractured at the base cross-section and the walls completely separated from the foundation (Fig. 12).

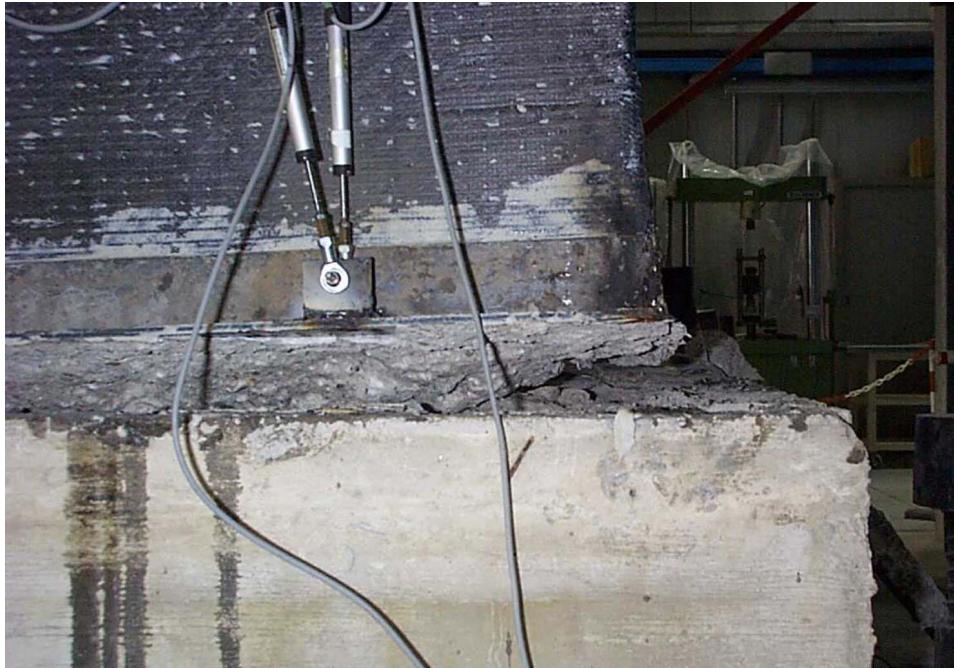


Fig. 12. Failure at the base of the L-shaped wall W1.

Then, the final cyclic test was performed; a series of three cycles at 200 mm has been applied, followed by three cycles corresponding to a displacement of 250 mm; the test ended with a final cycle at 300 mm. The global hysteresis loop is shown in Fig. 13. During the first three cycles no additional major damage was observed. As the test progressed with the next cycles, horizontal cracks were observed at the base of both columns. At many points throughout the structure the CFRP laminates started debonding from the surface of concrete and quite large cracks, mostly in beams, were noticed. Since the structure had already reached its maximum strength during the 1.5 test, no further increase of base shear forces could have been expected. Even though no increase of force capacity took place, the structure as a whole was able to sustain further cycles at increasing amplitudes without noticeable drops in strength and maintained rather good energy dissipation (see Fig. 13). In particular, the CFRP laminates proved to be able to prevent the spalling of concrete far beyond the flexural failure of the cross-sections.

6. Global performance of original versus repaired structure

The discussion presented in this section aims at comparing the global performance of original versus repaired frames in terms of displacements, strength, energy and frequency. It is underlined that comparisons are performed considering the behaviour of the two configurations under the same earthquake intensity. For this reason, the derived considerations about the effects of

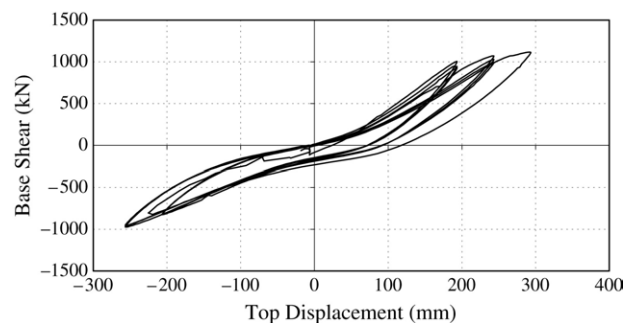


Fig. 13. Base shear–top displacement curves under the cyclic test.

the same accelerogram on the original and repaired frame include also the cumulative damage due to the previous tests.

The base shear–top displacement relationships under the design earthquake highlight that the displacement demand was larger for the repaired structure compared to that of the original; the repaired structure showed a lower stiffness compared to the original as demonstrated by the lower slope of the curve (Fig. 14). The maximum top displacement was about 75% larger in the repaired structure (Table 1); the FRP repair enabled the structure to withstand the increased displacement demand without significant damage and with an increase of maximum base shear equal to about 12%. A different behavior is observed with respect to the 1.5 design earthquake because the displacement demand was similar in the two configurations (Fig. 15). The original and repaired structures attained an almost equal maximum base shear and a maximum top displacement only about 8% larger for the repaired compared to the original structure (Table 1).

Table 1
Summary of global performance of original and repaired structure

Test	Max. top displacement (mm)	Max. base shear (kN)	Final total absorbed energy (kJ)	Initial frequency (Hz)	Final frequency (Hz)
1.0 original structure	100.69	1161.9	138.9	2.76	1.29
1.5 original structure	273.89	1470.4	345.1	1.29	0.78
1.0 repaired structure	175.91	1304.2	139.7	2.00	1.01
1.5 repaired structure	296.39	1457.9	347.9	1.02	0.77
Cyclic	293.44	1116.6	627.3	*	*

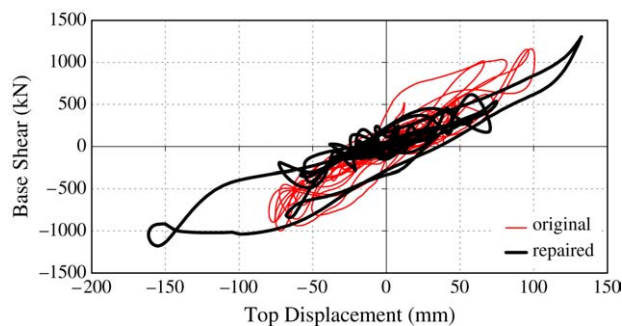


Fig. 14. Base shear–top displacement curves under the design earthquake.

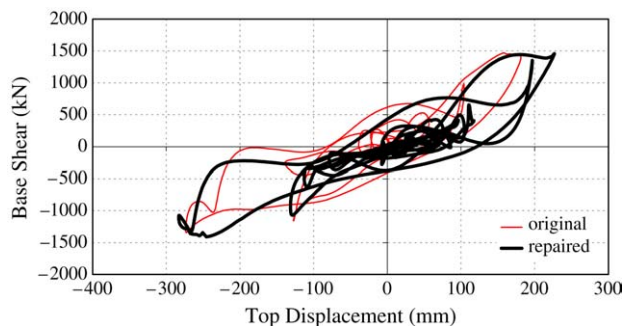


Fig. 15. Base shear–top displacement curves under the 1.5 design earthquake.

However, the total absorbed energy was more than 60% larger (Table 1). For both intensities, the progress of damage in the repaired structure resulted in a global symmetrical response (Figs. 14 and 15). Under the cyclic test the repaired structure was able to withstand a maximum top displacement almost equal to the maximum demand under the 1.5 design earthquake, even though its maximum base shear was about 76% of that under the 1.5 design earthquake.

By totalling up the areas under hysteretic cycles of shear–relative displacement relationships of each story, it was possible to obtain information about the energy dissipation at each story over time. In all tests, it was noted that the major energy dissipation occurred at the first level and that the contribution of other levels decreased as their height increase (Table 2); with respect to each

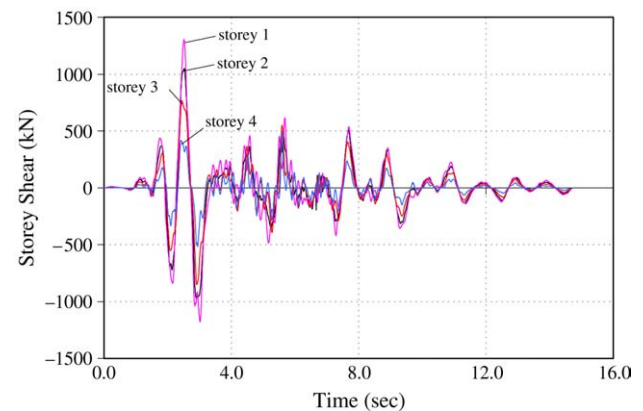


Fig. 16. Story shear–time curves at all stories of the repaired structure under the design earthquake.

story, the most significant amount of energy was dissipated during the first seconds of the accelerogram and this was reflected on trends of shear and relative story displacement over time by the presence of a very large initial cycle followed by all narrow cycles. An example concerning the repaired structure under the design earthquake is depicted in Figs. 16 and 17 where story shear–time and relative story displacement–time trends for the four stories are shown. By totalling up the instantaneous energy dissipated at all four stories, it was possible to obtain total dissipated energy–time relationships. Such curves highlight that the repaired structure was characterized by a faster dissipation compared to the original under the design earthquake; they reached the same amount of final absorbed energy after about eight seconds (Fig. 18 and Table 2). Under the 1.5 design earthquake, the hysteretic behaviour of original and repaired structures was indeed almost identical (Fig. 19).

The frequency values reported in Table 2 show that the damage suffered by the original structure during the design earthquake caused a reduction by about 50% of the first frequency (from 2.76 Hz to 1.29 Hz). A further decrease by about 40% (from 1.29 Hz to 0.78 Hz) of the first frequency corresponded to the application of the 1.5 design earthquake. The injection of cracks with epoxy resin enabled a partial recovery (i.e., about 50%) of the initial frequency of the

Table 2
Summary of story performance of original and repaired structure

Test		Story 1	Story 2	Story 3	Story 4
1.0 original structure	Drift (mm)	26.20	27.78	27.65	22.04
	Shear (kN)	1161.9	1000.6	872.4	561.0
	Energy (J)	54.9	42.4	29.5	11.9
1.5 original structure	Drift (mm)	99.94	72.72	59.94	47.02
	Shear (kN)	1470.4	1195.1	918.4	584.5
	Energy (kJ)	171.1	95.7	51.4	18.4
1.0 repaired structure	Drift (mm)	54.79	47.33	43.25	34.34
	Shear (kN)	1304.2	1049.3	852.4	516.8
	Energy (kJ)	64.3	39.9	25.1	10.5
1.5 repaired structure	Drift (mm)	98.77	78.37	66.95	54.16
	Shear (kN)	1457.9	1294.0	1015.5	593.7
	Energy (kJ)	170.0	95.0	58.5	25.1
Cyclic	Drift (mm)	94.70	78.03	67.28	59.69
	Shear (kN)	1116.6	1087.1	799.3	322.0
	Energy (kJ)	252.3	193.3	128.3	53.3

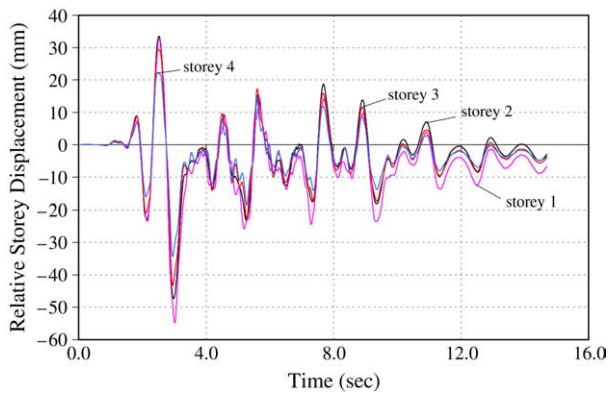


Fig. 17. Relative story displacement–time curves at all stories of the repaired structure under the design earthquake.

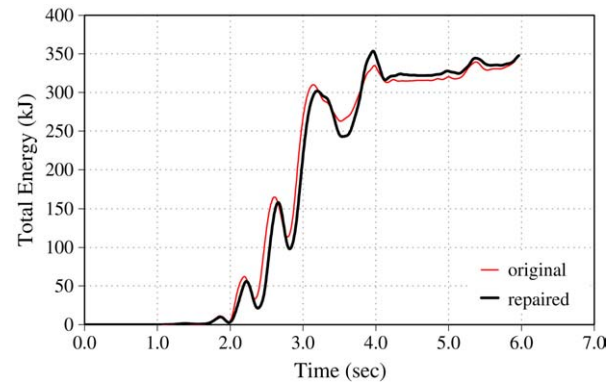


Fig. 19. Total dissipated energy under the 1.5 design earthquake.

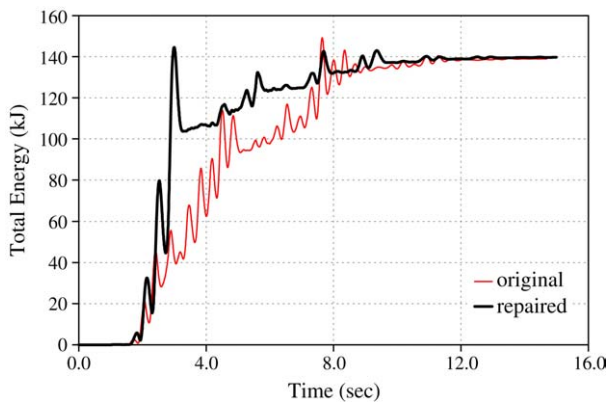


Fig. 18. Total dissipated energy under the design earthquake.

repaired frame (the frequency changed from 1.29 Hz to 2.00 Hz). As for the original structure, the design earthquake caused a reduction by about 50% of the frequency (from 2.00 Hz to 1.01 Hz). In spite of the different initial value, the frequency of the repaired structure after the application

of the 1.5 design earthquake resulted to be almost equal to the one characterizing the original frame.

7. Local performance of original versus repaired structure

The present section deals with local performance in terms of joint rotations and member deformation. From the joint rotation–time relationships under the design earthquake it was observed that, independently of the position (i.e., column or shear wall), all the joints of the repaired structure had maximum rotations larger than those of the original. Joints of both original and repaired structures exhibited low rotations until about 2 s and after about 10 s; those on the original showed peak rotations at about 4.5 and 7.5 s, whereas peak values on the repaired were recorded between 2 and 3 s. A typical example is provided in Fig. 20 with reference to the joint between beam B2 and columns C1 (Fig. 2) at first level. A different behaviour was noted under the 1.5 design earthquake with very similar maximum rotations of joints of original and repaired structures; for this intensity, the comparison can be made only in the first 6 s

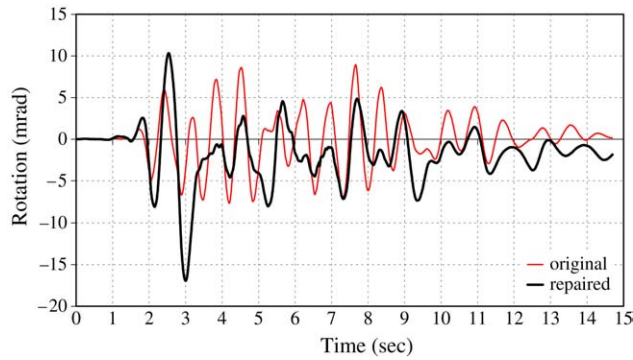


Fig. 20. Rotation-time of B2-C1 joint at first level under design earthquake.

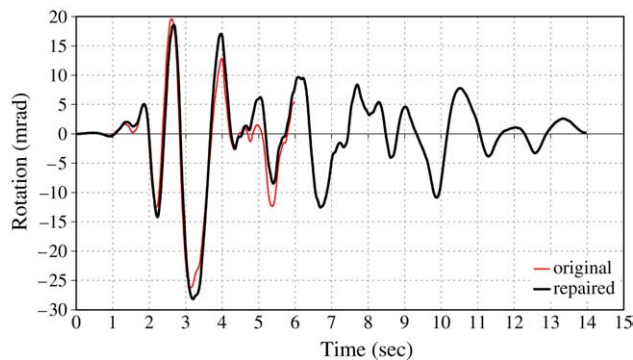


Fig. 21. Rotation-time of B2-C1 joint at first level under 1.5 design earthquake.

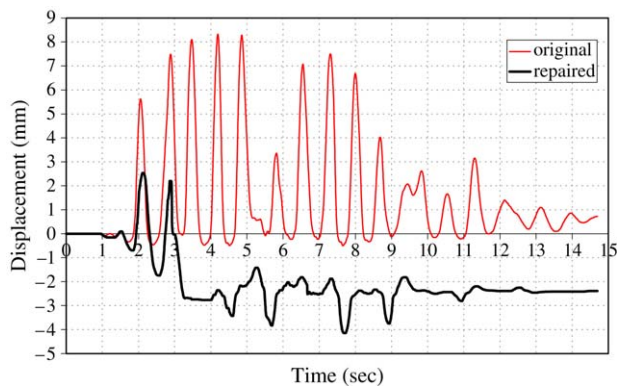


Fig. 22. Displacement-time for shear wall W1 at first story under design earthquake (diagonal 1).

as the test on the original frame was stopped at that time in order to avoid it becoming too damaged to be repaired. A typical example regarding the above B2-C1 joint at the first level is depicted in Fig. 21.

During the tests, information about the elongation/shortening of shear walls W1 and W2 (Fig. 2) along their diagonals were provided by displacement transducers. On both walls, a significant stiffening effect was measured at both first and second stories (diagonal displacement transducers were installed on the walls only at those

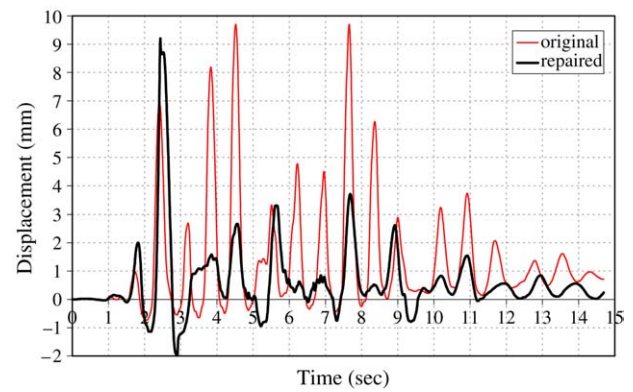


Fig. 23. Displacement-time for shear wall W2 at first story under design earthquake (diagonal 2).

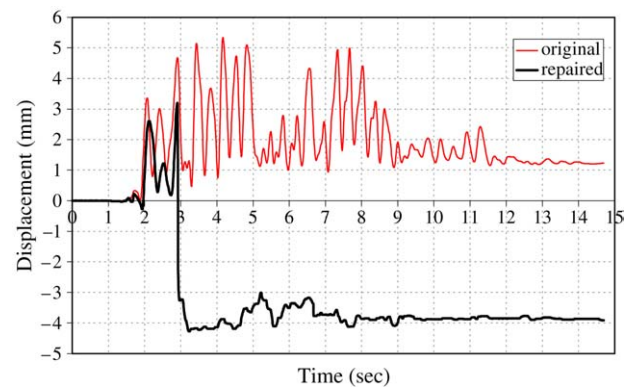


Fig. 24. Displacement-time for shear wall W1 at first story under design earthquake (diagonal 2).

two stories) due to the presence of the FRP laminates that reduced the diagonal deformations during the design earthquake; trends referred to the first story are depicted in Figs. 22 and 23 where elongations and shortenings are positive and negative, respectively. The almost constant displacement between about 3 and 4 s followed by small displacements on the repaired wall W1 (Fig. 22) was recorded also by the transducer placed on the same wall on the other diagonal (Fig. 24 referred to diagonal 2) and it was probably due to the opening of an horizontal crack at its base (Fig. 12). The stiffening effect of the FRP laminates became even more pronounced for both walls under the 1.5 design earthquake; typical displacement-time trends are depicted in Figs. 25 and 26 with reference to parallel diagonals placed on wall W2 at first and second story, respectively; as pointed out above, the comparison can be performed only up to the time when the 1.5 design earthquake was stopped on the original structure.

8. Conclusions

The results of pseudodynamic tests performed on an RC structure repaired with CFRP laminates have been presented and discussed in the paper. The experimental

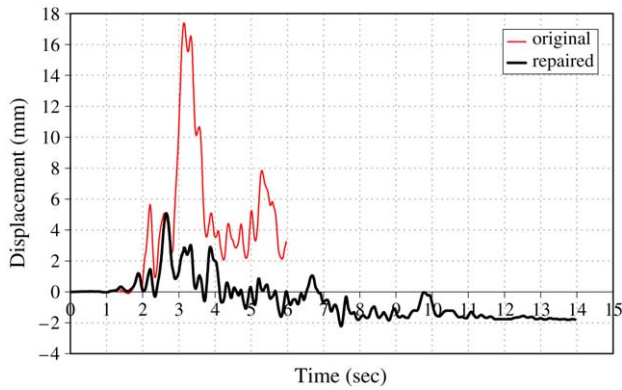


Fig. 25. Displacement–time for shear wall W2 at first story under 1.5 design earthquake (diagonal 1).

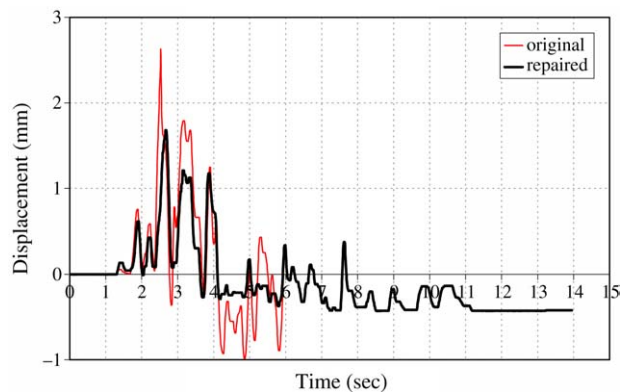


Fig. 26. Displacement–time for shear wall W2 at second story under 1.5 design earthquake (diagonal 1).

outcomes point out that the repaired structure did not attain any local mechanism above the foundation, but its failure was controlled by the column/wall-footing interface. The occurrence of such local crisis due to the termination of the CFRP laminates at the bottom of columns/walls did not allow the repaired structure to show the capacity increase that could be potentially attained due to the FRP strengthening. As observed in other tests [6,16], interface failures represent a typical issue of FRP composites that are generally unable to withstand high stress concentrations. The authors believe this is a critical aspect that still needs to be further investigated.

The main outcomes of the experimental campaign can be summarized as follows:

- (1) the experimental tests confirmed that CFRP composites allowed the repaired structure to withstand the displacement demand under both design and 1.5 design earthquakes with moderate damage above the foundation;
- (2) the repaired structure showed a large displacement capacity without exhibiting any loss of strength and was able to provide an energy dissipation very similar to that of the original configuration;

- (3) the cyclic behavior of the repaired structure was stable and no significant effect of cumulative damage was observed on the strengthened elements;
- (4) regarding the local deformability of joints and the shear walls: the rotations of CFRP repaired joints always overcame those of original joints under the design earthquake, whereas no significant difference was measured under the 1.5 design earthquake. The presence of CFRP laminates over the entire height of the shear walls reduced the deformability of such members at any earthquake intensity.

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